
Appendix A: Closed-form expression for the mean sum of squares of a gene's Pearson residuals with a Poisson model

For convenience, this is copied – with minor changes – from [1]. The formula is sparse – involving only entries of the UMI count matrix and row and column sums. Hence, the SSQ of genes' Pearson residuals can be computed sparsely. The dense Pearson residuals matrix is not needed.

We use the notations

X_{gc} = the UMI count for gene g in cell c

$G_g = \sum_c X_{gc}$ the total count for gene g

$\pi_{c|g} = X_{gc}/G_g$ the fraction of the total count of gene g that is in cell c

$D_c = \sum_g X_{gc}$ the total count for cell c – its sequencing depth

$T = \sum_{gc} X_{gc}$ the total of all counts in the matrix

C = the total number of cells in the matrix

$D = T/C$ the mean sequencing depth of the matrix

$\overline{D}_c = D_c/D$ the normalized sequencing depth of cell c

The Pearson residuals are

$$R_{gc} = (X_{gc} - \hat{\mu}_{gc}) / \sqrt{\hat{\mu}_{gc}}$$

with the maximum likelihood estimate

$$\hat{\mu}_{gc} = \frac{G_g \cdot D_c}{T}$$

Squaring gives

$$\begin{aligned} R_{gc}^2 &= \frac{(X_{gc} - \hat{\mu}_{gc})^2}{\hat{\mu}_{gc}} \\ &= \frac{X_{gc}^2}{\hat{\mu}_{gc}} - 2 \cdot X_{gc} + \hat{\mu}_{gc} \end{aligned}$$

Summing over all cells c gives S_g , the sum of squares of Pearson residuals for gene g :

$$\begin{aligned}
S_g &= \sum_c \left(\frac{X_{gc}^2}{\hat{\mu}_{gc}} - 2 \cdot X_{gc} + \hat{\mu}_{gc} \right) \\
&= \sum_c \left(\frac{X_{gc}^2}{\hat{\mu}_{gc}} \right) - 2 \cdot G_g + G_g \quad \text{because } \sum_c X_{gc} = \sum_c \hat{\mu}_{gc} = G_g \\
&= \sum_c \left(\frac{X_{gc}^2}{\hat{\mu}_{gc}} \right) - G_g \\
&= \sum_c \left(\frac{X_{gc}^2}{\frac{G_g \cdot D_c}{T}} \right) - G_g \quad \text{using } \hat{\mu}_{gc} = \frac{G_g \cdot D_c}{T} \tag{*} \\
&= \left(\sum_c \frac{\pi_{c|g} \cdot X_{gc}}{D_c} \right) \cdot T - G_g \quad \text{since } \pi_{c|g} = X_{gc}/G_g
\end{aligned}$$

Dividing by the number of cells in the UMI matrix, C , gives the mean sum of squares of Pearson residuals for gene g :

$$\begin{aligned}
M_g &= \left(\sum_c \frac{\pi_{c|g} \cdot X_{gc}}{D_c} \right) \cdot \frac{T}{C} - \frac{G_g}{C} \\
&= \left(\sum_c \frac{\pi_{c|g} \cdot X_{gc}}{D_c} \right) \cdot D - \frac{G_g}{C} \quad \text{because } T/C = D \\
&= \left(\sum_c \pi_{c|g} \cdot \frac{X_{gc}}{D_c/D} \right) - \frac{G_g}{C} \\
&= \left(\sum_c \pi_{c|g} \cdot \frac{X_{gc}}{\overline{D}_c} \right) - \frac{G_g}{C} \quad \text{since } \overline{D}_c = D_c/D - \text{the normalized sequencing depth of cell } c
\end{aligned}$$

Observations:

- The summation includes *only* cells on which gene g has nonzero counts
- The mean value of the denominators $\overline{D}_c$ over all cells in the UMI count matrix is 1
- The subtracted term $\frac{G_g}{C}$ is negligible for lowly expressed genes
- Since $\sum \pi_{c|g} = 1$, the summation is the convex combination of the ratios $X_{gc}/\overline{D}_c$. This provides intuition for the fact that many values of M_g are near 1. They are convex combinations of terms that are often close to 1: most nonzero counts X_{gc} equal 1 and the mean of the $\overline{D}_c$ equals 1

Appendix B: Closed-form expression for a cell's contribution to the sum of squares of a gene's Pearson residuals

Continuing the above derivation from equation (*) identifies each cell's contribution to S_g

$$\begin{aligned} S_g &= \sum_c \left(\frac{\frac{X_{gc}^2}{G_g \cdot D_c}}{\frac{T}{T}} \right) - G_g \\ &= \sum_c \left(\frac{X_{gc}^2 \cdot T}{G_g \cdot D_c} - X_{gc} \right) \quad \text{because } G_g = \sum_c X_{gc} \\ &= \sum_c \left(\frac{X_{gc}}{G_g} \cdot \frac{T}{D_c} - 1 \right) \cdot X_{gc} \end{aligned}$$

Denoting the contribution of cell c to S_g as K_{gc}

$$\begin{aligned} K_{gc} &= \left(\frac{X_{gc}}{G_g} \cdot \frac{T}{D_c} - 1 \right) \cdot X_{gc} \\ &= \left(\left(\frac{X_{gc}}{G_g} \right) / \left(\frac{D_c}{T} \right) - 1 \right) \cdot X_{gc} \end{aligned}$$

Observations:

- The numerator X_{gc}/G_g is the fraction of the count of gene g in cell c
- The denominator D_c/T is the fraction of the total UMI count in cell c
- Hence, the ratio compares the contribution of cell c to the count of g with the cell's contribution to the matrix's total count
- If this ratio equals 1, there is *no* contribution to the SSQ of the PR of g
- If it is less than 1, the cell's contribution to the SSQ of the PR is *negative*

Reference

- [1] Victor Klebanoff. Normalization and gene selection for single-cell rna-seq umi data using sampling-adjusted sums of squares of pearson residuals with a poisson model. *bioRxiv*, pages 2023–12, 2023.